TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	BEOL-compatible nitride electronics enabled by 2D materials-assisted epitaxy			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Electrical・Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	Compound Semiconductor Devices	Electrical and Electronic Components	Ultra-High-Frequency Generation Devices
Concept of Proposed Technology	We propose a BEOL integration technology that enables GaN HEMT modules to be co-integrated with Si CMOS platforms to add high-efficiency power functions without consuming valuable FEOL area or violating the post-CMOS thermal budget. The key advantage is to bring high current handling and high breakdown switching capability of wide-bandgap semiconductors to the BEOL, addressing power-delivery and routing bottlenecks that increasingly limit advanced-node CMOS systems. The key enabler is 2D material-assisted epitaxy. sp^2-bonded 2D interlayers possess a low-diffusion-barrier and atomically smooth growth front that expands the nitride growth window toward ≤400 °C, while decoupling lattice registry and accommodating severe mismatch for nitride-on-Si integration. In addition, the 2D interlayer provides a weakly coupled, release-ready interface, which enables membrane lift-off and low-temperature bonding. Using this platform, we pursue two complementary routes: direct BEOL-compatible growth of AlGaN/GaN HEMT heterostructures on 2D templates on CMOS wafers, and transfer-enabled low-temperature integration of donor-grown, high-performance GaN HEMT layers for high-current applications.			

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Rationale for Joint R&D Project	This joint R&D project is required to establish a manufacturable pathway for BEOL co-integration of GaN HEMT modules with Si CMOS, which is currently constrained by the strict post-CMOS thermal budget, stringent contamination control and materials allow-lists for device fabrication, and the lattice mismatch between nitrides and completed FEOL stacks. Overcoming these barriers requires a tightly coupled workflow in 2D interlayer engineering, ultralow-temperature nitrides growth, heterointerface control for HEMT channels, BEOL-compatible device processing, and wafer-level integration validation. Collaboration is therefore necessary: the PI team contributes the 2D material-assisted epitaxy and release/transfer technologies and device module development, while the Korean SME provides completed FEOL platforms, CMOS-compatible contamination control, and wafer-scale qualification required to demonstrate compatibility and scalability. The partnership enables closed-loop optimization from materials to devices to wafer-level integration, reducing technical and manufacturability risk.
Keywords	1. GaN-on-Si CMOS 2. GaN HEMT 3. BEOL transistor 4. 2D materials-assisted epitaxy

(max 10 pages)

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)

R&D Objectives

The objective of this project is to enable BEOL-compatible integration of nitride HEMT devices on Si by addressing the temperature and lattice-mismatch constraints that limit conventional nitride epitaxy.

At advanced nodes of Si CMOS technology, two classes of limitations are increasingly evident. On the device side, continued logic scaling requires stronger electrostatics and tighter control of process variations. This drives architectural evolution such as gate-all-around (GAA) and related designs, while longer-term efforts explore alternative channel materials including 2D semiconductors.[1, 2] In parallel, system-level constraints, such as power delivery, interconnect RC delay, thermal density, and integration complexity, have become key challenges. As a result, roadmaps give increased attention to interconnect and power-delivery co-optimization and the introduction of BEOL compatible functional modules to relieve routing and power-management bottlenecks without modifying the FEOL device stack. Representative examples include backside power delivery networks (BSPDN) and BEOL transistors.[3, 4]

BSPDN restructures the power delivery network by routing power through the wafer backside, which reduces supply-voltage drop and alleviates frontside routing congestion. By contrast, BEOL transistors are introduced to add local device functionality, such as power gating and level shifting, without consuming valuable FEOL area. In practice, BEOL transistors typically rely on poly-Si, a-Si, or oxide semiconductors as channel materials due to their low processing temperature.[5] However, the limited crystallinity and imperfect interfaces in these materials often lead to reduced driving current, higher contact resistance, and threshold-voltage instability, which constrain more demanding power-management use cases. Alternatively, III-nitride semiconductors are attractive because their wide bandgap and high breakdown capability enable low-loss switching with strong current handling. However, BEOL integration of nitrides on Si CMOS is particularly challenging because the completed interconnect stack poses a strict thermal budget and contamination constraints, which limits the process window for achieving device-level nitride materials and interfaces.

Motivated by this gap, we propose a BEOL-compatible nitride integration platform on Si CMOS enabled by 2D-material-assisted epitaxy. In this technique, the 2D interlayer can alleviate the need for strict lattice matching between the epilayer and the substrate meanwhile provide an atomically smooth growth front with low diffusion barrier (Fig. 1).[6] This can enable unconventional nucleation and lattice modulation, which can expand the process window toward BEOL-compatible temperatures while maintaining crystalline quality and is particularly attractive for highly mismatched nitride-on-Si integration. In addition, the weakly coupled interface offered by the 2D layer enables membrane release and transfer, providing a practical route to integrate as-grown nitride device layers onto target substrates through low-temperature processes. The PI has pioneered 2D-material-assisted epitaxy techniques, including remote epitaxy, quasi van der Waals (qvdW) epitaxy, and selective-area epitaxy.[7-10] By engineering the 2D-mediated growth interface, these methods can reduce the required growth temperature and introduce epitaxial modes that accommodate mismatch and decouple lattice registry from the substrate, which is essential for developing BEOL-compatible nitride integration on Si CMOS. Furthermore, the PI has demonstrated nitride-based acoustic and optoelectronic devices enabled by remote epitaxy and subsequent layer transfer onto target

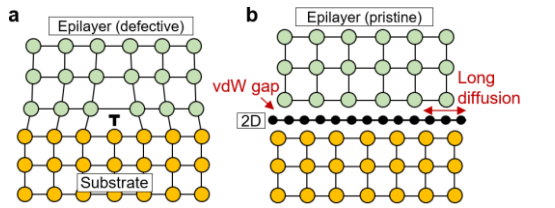

Fig. 1 2D-material-assisted epitaxy for low-temperature growth. (a) Conventional direct epitaxy suffers from limited adatom diffusion and interfacial reactions. (b) 2D interlayer lowers the adatom diffusion barrier and enables long-range diffusion

substrates. Notable examples include micro-LED platforms with record pixel density and GaN acoustic devices with high electromechanical coupling.[9, 11] These demonstrations validate a low-temperature, non-destructive release-and-transfer route for nitride device layers, which potentially enables BEOL integration on Si with a completed FEOL stack.

Building on this foundation, we will develop a BEOL-compatible nitride HEMT module on Si using two complementary pathways. First, direct BEOL-compatible growth: we will grow nitride films

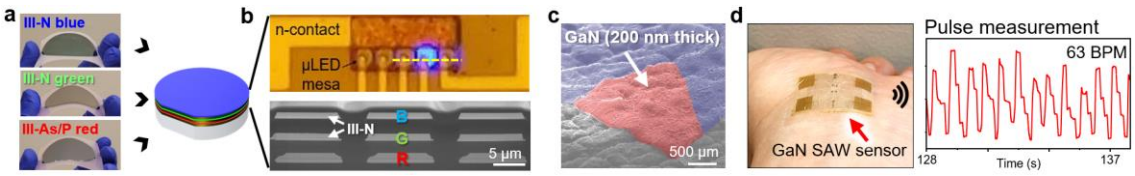

Fig. 2 Record-breaking nitride devices enabled by 2D materials-assisted epitaxy. (a) Exfoliated LED layers. (b) Vertically stacked micro-LEDs with the world's highest pixel density. (c) Ultrathin GaN membrane can be conformally integrated onto textured surfaces. (d) Multi-modal SAW sensors with high sensitivity.

on 2D templates confined to dielectric regions on silicon. The primary objective is to achieve nitride formation and integration within a BEOL-compatible temperature window, while aiming to maintain device-level defect and stress without relying on conventional high-temperature nitride epitaxy. Second, transfer-enabled integration: for device variants that require higher current drive and tighter performance margins, we will form high-quality nitride device layers on suitable growth templates and then use the weakly coupled 2D interface to enable membrane release and low-temperature bonding onto Si without disturbing the FEOL stack.

R&D Target Specifications

BEOL-integrated III-nitride transistors by direct growth on completed Si CMOS: The growth temperature will be strictly controlled under the post-CMOS thermal budget (400 °C). The growth will be performed in MOCVD using H_2, TMGa, TMAl, and NH_3 on 2D materials including graphene, h-BN, and their amorphous counterparts, with CMOS-compatible contamination control. Considering the intended application of BEOL compatible GaN HEMT is power routing/power gating, the goal is to demonstrate material Hall mobility ≥150 $cm^2 \cdot V^{-1} \cdot s^{-1}$ and device I_{on} ≥200 μA/μm, threshold-voltage drift ≤150 mV after 10^4 s. While subthreshold swing and I_{off} is not the dominant requirements for power switches, therefore we target SS ≤200 mV/dec and I_{off} ≤20 nA/μm at room temperature.

Wafer-scale transfer and bonding of a GaN HEMT layer onto Si CMOS: Because the nitride layer can be grown on an optimized substrate at an optimized temperature prior to integration, the performance goals are to demonstrate material Hall mobility ≥1500 $cm^2 \cdot V^{-1} \cdot s^{-1}$, and device I_{on} ≥0.5 A/mm, threshold-voltage drift ≤100 mV after 10^4 s. We further target SS ≤ 100 mV/dec and I_{off} ≤ 1 nA/μm at room temperature. The key integration targets for this approach are low-temperature bonding and transfer-defect control; therefore, the maximum bonding/anneal temperature will be limited to ≤ 400 °C, and wafer-scale integration will be demonstrated on ≥100 mm substrates.

R&D Plan

Task 1. Ultralow temperature nitride growth enabled by 2D material assisted epitaxy

The goal here is to establish a repeatable ≤400 °C MOCVD growth process for device-grade nitride layers on 2D interlayers, which involves controlled nucleation, coalescence, and formation interfaces suitable for transistor channels.

Task 1.1 Identify the most suitable 2D interlayer for direct nitride growth on completed Si CMOS

Because post-FEOL synthesis of single-crystalline graphene and h-BN is thermodynamically and process-wise challenging under a ≤400 °C BEOL budget, our primary path is to enable ultralow-temperature nitride epitaxy on BEOL-compatible sp^2-bonded 2D templates, including amorphous graphene (a-Gr) and amorphous BN (a-BN). The PI has pioneered the growth of these materials and has recently demonstrated nitride-based LED growth on such templates.[9, 10] Building on this foundation, we will systematically evaluate how these amorphous 2D materials modulate the nitrides growth, from the nucleation to lattice registration, and co-optimize 2D

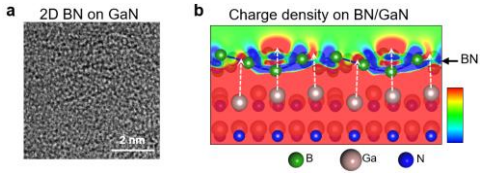

Fig. 3 a-BN assisted nitrides growth. (a) SEM image of the a-BN. (b) a-BN assisted GaN growth.

materials growth and nitride epitaxy. If nitride epitaxy on these 2D templates is not ideal, as a risk mitigation plan, we will consider utilizing dry-transferred single-crystalline graphene for nitride growth with balancing the lattice guidance and contamination control.

Task 1.2 Low-temperature nitrides growth on 2D materials

We will exploit the reduced adatom diffusion barrier on 2D surfaces, which can increase the effective surface diffusion length and thereby enable nitride nucleation and coalescence at substantially lower temperatures than on conventional 3D substrates. Prior reports suggest that long adatom diffusion lengths on 2D interlayers can broaden the growth window for III-nitrides and, under appropriately engineered conditions, yield comparable crystalline quality at reduced temperature. To fully leverage this expanded window, we will implement non-conventional MOCVD growth modes, including flow-modulated epitaxy and metal-modulated epitaxy, to decouple surface kinetics from gas-phase chemistry, control surface coverage, and maximize adatom migration time, all of which are expected to promote lateral growth and early coalescence under a ≤400 °C constraint. As a risk-mitigation strategy, we will also investigate post-nucleation thermal treatments within the BEOL budget to improve film quality by enabling defect reorganization prior to full coalescence.

Task 1.3 Interface control for device formation

A central challenge for ultralow-temperature nitride HEMT growth is achieving abrupt, low-trap heterointerfaces suitable for transistor channels. To address this, we will (a) optimize the GaN underlayer to provide a smooth, continuous template prior to barrier deposition; (b) employ growth modes that maximizes effective adatom migration time without exceeding the thermal budget, including flow-modulated epitaxy, growth interrupts, and pressure/V/III ratio tuning to promote step-flow–like lateral growth during the critical heterointerface formation; and (c) implement abrupt-composition switching at the interface by tightly controlling precursor or applying pre-exposures (e.g., NH_3/H_2 surface conditioning) to reduce intermixing while maintaining BEOL compatibility.

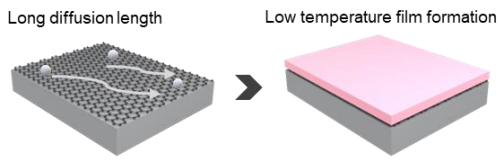

Fig. 4 Enhanced adatom diffusion on 2D materials promotes low-temperature nitride film formation.

Task 2 HEMT demonstration on 2D material assisted epitaxy enabled nitride heterostructures

The goal here is to translate the nitride heterostructures developed in Task 1 into functional HEMT/MIS-HEMT devices using a BEOL-compatible device fabrication flow, including low-temperature gate-stack and passivation formation, etch-damage control, and contact/access resistance engineering.

Task 2.1 Device architecture and BEOL-compatible process flow

We will select a transistor architecture aligned with power routing/power gating, prioritizing strong channel control and low conduction loss under the available thermal budget. The process will include (a) mesa definition and access-region engineering with minimized etch-induced damage, (b) BEOL-compatible dielectric deposition and interface conditioning for gate-stack formation, and (c) surface passivation to suppress trapping and stabilize device operation. Throughout this process, we will enforce a materials set and processing sequence consistent with post-CMOS constraints and will document a standardized test structure set (Hall bars, capacitor structures, etc.) to enable quantitative extraction of device and interface metrics.

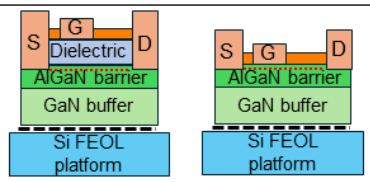

Fig. 5 Post-CMOS formation of nitride HEMT/MISHEMT devices.

Task 2.2 Contact/access resistance engineering for low-loss switching

For practical power switches, the conduction loss is often dominated by contact and access resistance. To address this, we will develop BEOL-compatible Au-free metallization and contact-formation strategies within the ≤400 °C budget by (a) optimizing Au-free contact-stack families (e.g., Ti/Al, TiN/TiW/Ta/W) and (b) low-thermal-budget surface preparation and interface engineering to enhance tunneling and reduce the effective barrier within the allowed anneal window. The access regions will be co-optimized to minimize access resistance without compromising leakage or isolation by (a) designing source-gate and gate-drain spacing, (b) electric-field management, and (c)

passivation strategies that suppress surface trapping.

Task 2.3 Power-switch performance validation and reliability

Device performance will be quantified using I_{on}, R_{on}, I_{off}, and SS, with standardized extraction structures and protocols. To ensure power-switch relevance, we will evaluate stability and trap-related degradation using (a) bias-stress testing with $|\Delta V_{th}|$ tracked up to 10^4 s and (b) pulsed/transient characterization to quantify dynamic R_{on}. The measured signatures will be used to guide process refinements within Task 2 and to provide structured feedback to Task 1 when heterointerface quality limits stability.

Task 3 Integrating nitride-based HEMT on completed Si CMOS platform

The goal here is to integrate the material and device modules in Tasks 1, 2 onto completed Si CMOS platform through collaboration with a Korean SME.

Task 3.1 FEOL integration setup

We will work with the Korean SME to define the FEOL starting platform, including wafer format, top-level materials and topography, allowed materials list, and maximum thermal exposure. Based on these constraints, we will establish an integration protocol that prevents cross-contamination and protect the completed CMOS stack. A split-lot plan will be implemented: (a) control wafers, (b) partial-module wafers with 2D interlayer formation/2D interlayer and nitrides formation, and (c) full-module wafers after the HEMT device formation, enabling clear attribution of any performance changes to specific integration steps.

Task 3.2 FEOL-integration demonstration

We will execute the BEOL nitride integration on FEOL wafers using the process established in Task 1, 2. The primary path is direct ultralow-temperature growth on 2D interlayers, followed by BEOL-compatible device fabrication and electrical benchmarking. Wafer-level metrics, uniformity, yield, and leakage/isolation, will also be tracked to assess compatibility with realistic FEOL topography and materials. Considering the direct-growth path may not be suitable for higher-power, high-current-density switching due to limited device performance, we will develop the wafer-scale transfer and low-temperature bonding, decoupling nitride material quality from post-CMOS thermal limits. In this route, device-grade nitrides are grown on optimized donor substrates and integrated onto FEOL using ≤400 °C bonding/anneal with void/crack control.

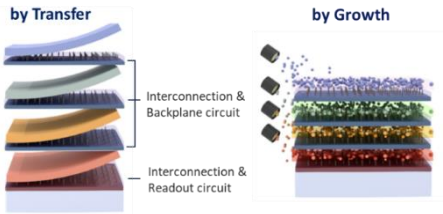

Fig. 6 Nitrides integration on Si CMOS enabled by direct growth or transfer.

Below, we provide a list of tasks, subtasks, and milestones. We also outline detailed, measurable deliverables for each milestone and specify the timeline for their completion within the overall project schedule.

Milestone 1.1 Identify BEOL-compatible 2D interlayers that survive MOCVD ambient and provide a uniform template for nitride nucleation. Deliverable: Raman spectra confirming 2D layer presence and no visible delamination/cracking after representative MOCVD ambient exposure.

Milestone 1.2 Demonstrate controlled nitride nucleation on selected 2D interlayers at low temperature (≤400 °C). Deliverable: SEM/AFM nucleation maps confirming nucleation coverage ≥90% with controllable island density/size distribution.

Milestone 1.3 Achieve continuous GaN film on 2D interlayers within the BEOL thermal budget. Deliverable: Continuous film with thickness discrepancy ≤10% confirmed by SEM/AFM. Dislocation density $\leq 10^{12}$ cm^{-2} characterized by XRD, SEM etch pits, or TEM.

Milestone 1.4 Demonstrate BEOL-compatible growth of AlGaN/GaN heterostructures with an interface suitable for HEMT channels under ≤400 °C constraints. Deliverables: AFM roughness confirming the surface roughness ≤1 nm. Cross-sectional TEM and EDS confirming an abrupt heterointerface. Hall measurement confirming the mobility ≥150 $cm^2 \cdot V^{-1} \cdot s^{-1}$.

Milestone 2.1 Define a HEMT architecture for power routing/power gating and establish baseline BEOL-compatible process modules for gate stack formation, passivation, and Au-free contacts. Deliverables: Finalized mask/layout, baseline process flow, and documented test structures (Hall, C-V) ready for fabrication.

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Milestone 2.2 Fabricate functional HEMT/MIS-HEMT devices using the baseline BEOL flow and demonstrate repeatable gating and power-switch-relevant metrics. Deliverables: Transfer I-V confirming I_{on} ≥200 μA/μm, I_{off} ≤20 nA/μm, and SS ≤200 mV/dec at room temperature, under the standardized protocol.

Milestone 2.3 Quantify stability and trap-related degradation and close the loop to process refinement. Deliverables: |ΔVth| ≤150 mV after 10^4 s under application-relevant bias-stress. Dynamic R_{on} from pulsed/transient tests.

Milestone 2.4 Establish a high-quality III-nitride heterostructure and HEMT/MIS-HEMT baseline on optimized donor substrates, providing a verified starting point for wafer-scale transfer and low-temperature bonding. Deliverables: On wafer Hall mobility μ ≥1500 $cm^2 \cdot V^{-1} \cdot s^{-1}$. Transfer I-V confirming I_{on} ≥0.5 A/mm, I_{off} ≤1 nA/μm, and SS ≤100 mV/dec at room temperature, under the standardized protocol. ΔVth| ≤100 mV after 10^4 s under application-relevant bias-stress.

Milestone 3.1 Define the FEOL starting platform with the Korean SME. Deliverables: Agreed FEOL starting platform spec (wafer format, top stack, allowed materials, and temperature tolerance).

Milestone 3.2 Demonstrate direct-growth BEOL nitride module on FEOL wafers. Deliverables: Wafer uniformity based on thickness/sheet-resistance mapping with ≤20% variation. Yield as the fraction of functional devices exhibiting repeatable transfer/output characteristics ≥70% of tested sites. Report wafer-level metric distributions (median and spread) for I_{on}, I_{off}, SS, and Vth, and include a representative stability tracking |ΔVth| up to 10^4 s using the standardized protocol.

Milestone 3.3 Wafer-scale transfer of donor HEMT and low-temperature bonding for high-current requirements. Deliverables: Demonstrate wafer-scale transfer and bonding of donor-grown nitride/HEMT layers onto FEOL wafers with bonding/anneal temperature≤400 °C and ≥100 mm format. Achieving controlled transfer defects with void and cracking area fraction ≤5% over mapped regions. Post-transfer, I_{on} retention ≥70% while maintaining acceptable stability trend.

Table 1 Project schedule and milestones.

Task	Year 1				Year 2				Year 3			
	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
1.1 Identify the most suitable 2D interlayer		M1.1										
1.2 Low-temperature nitrides growth			M1.2		M1.3							
1.3 Interface control for device formation							M1.4					
2.1 BEOL-compatible device architecture					M2.1							
2.2 Contact/access resistance engineering								M2.2				
2.3 Performance validation and reliability								M2.3	M2.4			
3.1 FEOL integration setup								M3.1				
3.2 FEOL-wafer demonstration											M3.2	M3.3

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	UIUC team will be responsible for: (a) developing and optimizing the 2D-assisted nitride epitaxy approach under the post-CMOS thermal budget; (b) establishing and demonstrating BEOL-compatible GaN HEMT/MIS-HEMT device modules, including gate-stack/passivation integration and contact/access resistance engineering; (c) performing materials and device characterization, performance benchmarking, and stability screening based on standardized protocols; and (d) providing an integration-ready process package (process flow, parameter windows, and acceptance/test protocols) to support FEOL-wafer demonstration with the Korean SME partner. UIUC team will provide the research personnel, facilities access, and project coordination needed to execute the planned tasks and milestones.
Expected R&R of Korean SME Partner	We expect the Korean SME partner to serve as the CMOS integration authority and manufacturing validation of BEOL-compatible nitride HEMT modules on completed Si CMOS wafers. Beyond template supply and fabrication support, the SME will define integration constraints, execute wafer-level qualification, and establish manufacturability benchmarks required for TRL progression from laboratory-scale demonstration (TRL 3/4) to prototype-level validation (TRL 5/6). **1. Single crystal 2D Template on amorphous surface for BEOL Integration** The Korean SME could develop and supply confined quasi–van der Waals (qvdW) platforms consisting of single-crystalline or highly oriented 2D materials formed on amorphous dielectric-coated Si substrates. This platform will include: • Si wafers coated with BEOL-representative amorphous dielectric layers • Patterned dielectric regions defining confined geometries • Selectively formed 2D layers (e.g., graphene, h-BN, MoS_2 or related materials) within confined dielectric pockets The confined architecture serves as a geometry-defined growth window, enabling: • Spatially localized nitride nucleation • Suppression of parasitic lateral overgrowth • Stress confinement within dielectric regions The SME will ensure: • Wafer-scale dimensional uniformity of confined regions • Statistical mapping of 2D crystalline quality • Reproducibility of amorphous dielectric/2D interface properties This confined dielectric/2D platform provides a CMOS-representative foundation for evaluating BEOL-compatible nitride integration without requiring direct access to advanced production FEOL wafers. . **2. Subsequent BEOL-Compatible Deposition Support (Low-Temperature Growth, <400oC)** The Korean SME could support and enable subsequent low-temperature nitride deposition processes to ensure compatibility with BEOL constraints. Responsibilities include: • Preparation of dielectric-coated Si substrates capable of withstanding MOCVD or related growth environments within ≤400 °C thermal limits • Verification of dielectric and 2D layer stability under representative growth ambient exposure • Definition of practical thermal and materials compatibility boundaries relevant to BEOL processing • Support for split-platform validation (bare dielectric vs confined 2D platform) to isolate the role of the confined 2D interface in low-temperature nitride formation This role ensures that low-temperature growth is evaluated under realistic CMOS-relevant substrate conditions and that the confined 2D platform maintains structural integrity during deposition. **3. Packaging and System-Level Testing**

(max 10 pages)

	The Korean SME will lead backend-compatible integration and wafer-level evaluation necessary for TRL progression from materials demonstration to prototype-level validation. Responsibilities include: • Wafer-level device electrical characterization (DC, RF, breakdown voltage, leakage current, thermal stability). • High-frequency testing relevant to HEMT performance (e.g., fT, fmax, power handling capability). • Thermal reliability testing under BEOL-compatible operation conditions. • Accelerated lifetime testing to support reliability validation. • Failure analysis (where applicable) to identify integration-induced degradation mechanisms. Testing results will be systematically shared with UIUC to refine materials growth parameters and device design in iterative development cycles.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Successful BEOL integration of III-nitride transistors addresses a critical bottleneck in advanced semiconductor platforms: the growing need for efficient, distributed power-management elements without perturbing the FEOL device stack. This capability would strengthen the ecosystem for heterogeneous integration of wide-bandgap devices with Si CMOS and enable differentiated power-management solutions for high-performance computing, AI accelerators, and energy-efficient edge systems. Building on these drivers, the proposed joint R&D combines UIUC team's materials/device innovation with the Korean SME's integration and manufacturing capabilities to accelerate translation from laboratory demonstration to industrial adoption, while contributing to workforce development and supply-chain competitiveness in next-generation semiconductor technologies. **Expected outcomes** Key technical outcomes will progress from materials to devices to platform integration: (a) a repeatable ≤400 °C MOCVD process window for device-grade III-nitride growth on BEOL-compatible 2D interlayers with contamination-aware integration protocols; (b) demonstration of functional GaN HEMT/MIS-HEMT devices suitable for power routing/power gating with quantified performance and stability metrics, and (c) BEOL-integrated nitride HEMTs on completed Si CMOS, demonstrated at wafer scale with quantified uniformity/yield and key device-metric distributions under standardized protocols. **Commercialization potential** The project targets BEOL-compatible power-management functions that are increasingly important in advanced CMOS systems, including power gating, power routing, and localized switching. By enabling nitride-based switches within the BEOL thermal budget, the outcomes can reduce conduction loss and improve power-delivery efficiency compared with conventional BEOL transistor materials, potentially translating into better energy efficiency and higher effective compute performance at the system level. The collaboration with the Korean SME positions the technology for commercialization by providing a realistic pathway to wafer-level integration, interconnection, packaging, and validation, and by generating manufacturing-relevant deliverables that can be adopted for pilot production and customer engagement.

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